

Verification Continuum™

**AEware Application Note
for
compute_formal_core_all**



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AE-ware Application Note for compute_formal_core_all

This document describes the purpose and usage of the “compute_formal_core_all” AE-ware.

For product support, see <https://www.synopsys.com/support.html>.

Introduction

While computing formal core, the user would like to consider all FPV proven properties as well as all the inconclusives FPV properties at their respective safe depths.

Currently, the in-built tool commands either allows the user to consider only the proven properties or proven+inconclusive at a min depth (the min depth gets applied to proven also, which the user would not want). In order to work around this limitation, the user needs to code a set of tcl commands to achieve a more comprehensive formal core.

This AE-ware provides this capability. This command allows the user to compute a comprehensive formal core by using the proven properties first, and then using the inconclusives properties at their respective depths.

This command also allows the user to bucket the inconclusive properties into unique depth buckets for faster runtime.

Prerequisites

- 1) VC Formal version needs to be 2021.09-1 or higher.
- 2) The availability of FPV results that is, the FPV run should have been performed before moving onto formal core analysis.

Location

`$VC_STATIC_HOME/auxx/ctg/tcl/library/proc/compute_formal_core_all.tbc`

Command Help

```
vcf>compute_formal_core_all -help
```

Usage: compute_formal_core_all # Compute Formal Core for both proven and inconclusive properties from FPV run. For inconclusive properties, analysis is done based on unique safe depth buckets or per property bases

[-max_time_proven] (Max time to be spent on proven FPV properties (Default: 1H))

[-max_time_inconcl] (Max time to be spent on inconcl FPV properties (Default: 1H))

[-mode_for_inconcl] (Values: 0 (Default. Unique safe depth buckets), 1 (Each property). Only applicable for inconclusives.)

Usage

The following steps need to be followed to use this AE-ware :

1. Run check_fv for FPV.
2. Source the AE-ware script.
3. Execute “compute_formal_core_all” command.

Example

```
vcf>.....
```

```
#run FPV
```

```
vcf> check_fv
```

```
#source AEware
```

```
vcf>Source
```

```
$::env($VC_STATIC_HOME)/auxx/ctg/tcl/library/proc/compute_formal_core_all.t  
bc
```

```
vcf>compute_formal_core_all -max_time_proven 30M -max_time_inconcl 5M -  
mode_for_inconcl 0
```

(In the above example we are setting max time for proven as 30 minutes and for each inconclusive unique safe depth bucket as 5 minutes)

Sample AE-ware output

```
371 vcf>source compute_formal_core_all.tbc
372 vcf>compute_formal_core_all -max_time_proven 30M -max_time_inconcl 5M -mode_for_inconcl 0
373 [AEware] Starting Comprehensive Formal Core Analysis
374 [AEware] WARNING : Liveness properties are not supported in Formal Core yet
375 [AEware] Identifying all the properties to be used for Formal core

493 [AEware] The unique safe depths for FTA are : 9 10 11 12 14 15 18 20 30 102
494 [AEware] Pre-processing complete
495 [AEware] Starting step-by-step Formal Core run
496 [AEware] STEP-1 : Formal Core run with proven FPV properties started
497 [Info] CFC_RESOURCE_MSG: Formal core computation can take time, please use fml_max_time to manage the allotted time.

506 [AEware] STEP-1 : Formal Core run with proven FPV properties complete
507 [AEware] STEP-2 : Formal Core run with inconclusive FPV properties started
508 [AEware] Mode 0 selected : Running based on unique depth buckets
509 [AEware] Bucket 0 Unique depth : 9
```

Known Limitations

Currently, Liveness properties are not supported for formal core analysis, hence only the safety properties are filtered in this AE-ware.

References

Please refer to *Section 19.7 VC Formal User Guide* for more info on formal core analysis in VC Formal.